

Hormone Lab Reference Ranges

Estrogen, testosterone, progesterone, DHEA, cortisol, thyroid — optimal vs conventional ranges for BHRT Danielle Alcala · nopriorauthorization.com

FEMALE HORMONE REFERENCE RANGES					MALE HORMONE REFERENCE RANGES			
Hormone	Conventional Range	Optimal (BHRT)	Timing of Draw	Notes	Hormone	Conventional Range	Optimal (TRT)	Notes
Estradiol (E2)	Follicular: 30–120 pg/mL	100–200 pg/mL (on therapy)	Day 19–21 of cycle (if cycling)	Serum preferred. Saliva used by some providers. Check per delivery method	Total Testosterone (M)	270–1070 ng/dL	600–1000 ng/dL (on therapy)	Morning fasting draw. Levels naturally highest 8–10am
	Luteal: 70–300 pg/mL				Free Testosterone (M)	9–30 pg/mL	15–30+ pg/mL (on therapy)	Critical measure — SHBG affects bioavailability of total T
Menopause:	<30 pg/mL							
Progesterone	Follicular: 0.1–0.9 ng/mL	2–20 ng/mL (on therapy)	Day 19–21 of cycle	Low progesterone = estrogen dominance symptoms. Must oppose estrogen	Estradiol (E2) — M	10–40 pg/mL	20–30 pg/mL (on therapy)	Testosterone aromatizes to E2. Too high = gynecomastia, water retention, mood
	Luteal: 2–25 ng/mL							
Menopause:	<0.1 ng/mL							
Total Testosterone (F)	15–70 ng/dL	50–100 ng/dL (on therapy)	Morning fasting	Many women therapeutic in 50–80 range. Watch for androgenic side effects	PSA	<4.0 ng/mL (age-dependent)	Baseline + q6mo on TRT	Must screen pre-treatment. Testosterone may increase PSA — monitor
Free Testosterone (F)	0.1–6.4 pg/mL	1.0–8.5 pg/mL (on therapy)	With total testosterone	Better indicator of active testosterone. SHBG affects total T	Hematocrit	38.3–48.6%	<52% on therapy	TRT increases RBC production. >54% = phlebotomy or dose reduction
DHEA-S (F)	35–430 mcg/dL (age-dependent)	150–250 mcg/dL	Any time	Declines with age. Pregnenolone precursor. Supports energy and libido	LH / FSH	Variable	Suppressed on TRT (expected)	Used to diagnose primary vs secondary hypogonadism pre-treatment
SHBG	18–114 nmol/L	Generally lower is better	With testosterone panel	High SHBG = low free T even if total T is normal				

ADRENAL & THYROID REFERENCE				PELLET DOSING REFERENCE – COMMON STARTING RANGES			
Hormone	Conventional Range	Optimal	Notes	Patient	Testosterone Pellet	Estradiol Pellet	Re-pellet Timing
Cortisol (AM)	6–23 mcg/dL	12–18 mcg/dL (8–9am draw)	Draw 8–9am exactly. PM cortisol naturally lower. Low = adrenal fatigue concern	Female — low libido, fatigue	50–100mg T	6–12.5mg E2 (if needed)	Every 3–4 months
DHEA-S (M)	80–560 mcg/dL (age-dep)	200–350 mcg/dL	Declines ~10% per decade after 30. Supports energy, libido, immune function	Female — post-menopause symptomatic	75–150mg T	12.5–25mg E2	Every 3–4 months
TSH	0.4–4.5 mIU/L	0.5–2.0 mIU/L (optimal)	Many functional practitioners target lower range. Symptoms matter as much as numbers	Male — mild hypogonadism	600–900mg T	Not typically	Every 4–6 months
Free T4	0.8–1.8 ng/dL	1.1–1.5 ng/dL	Prohormone. Converts to active T3. Low despite normal TSH = conversion issue	Male — significant deficiency	900–1200mg T	Not typically	Every 4–6 months
Free T3	2.3–4.2 pg/mL	3.2–4.2 pg/mL	Active thyroid hormone. Low T3 = hypothyroid symptoms even with normal T4/TSH	Note	Dose varies widely by weight, activity, prior therapy, and lab response	—	Always re-check labs 4–6 wks post-pellet to assess levels
Reverse T3	9.2–24.1 ng/dL	<15 ng/dL	Blocks T3 receptors. Elevated in stress, illness, low calorie diets				
Thyroid antibodies (TPO/TgAb)	Negative	Negative	Positive = Hashimoto's — changes treatment approach				
KEY REMINDERS – HORMONE THERAPY							
★ Always treat to SYMPTOMS + LABS together — never labs alone. A patient with optimal numbers who still feels terrible needs reassessment.							
★ Mammogram current before initiating estrogen therapy. PSA baseline before initiating testosterone in men.							
★ Progesterone must be used in women with an intact uterus receiving estrogen — unopposed estrogen increases endometrial cancer risk.							
★ Pellet therapy is IRREVERSIBLE once inserted — cannot be removed. Side effects must be managed until pellet dissolves (3–6 months).							
★ Hematocrit must be monitored in all men on TRT. Phlebotomy or dose reduction if >54%.							